

Lambert Leong, PhD

Senior AI/ML Scientist and Engineer — Health AI

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Summary of Qualifications

- PhD in Bioengineering with an AI/ML focus and an MS in Computer Science.
- 8+ years building and deploying health AI, machine learning, and biomedical research systems from model development through evaluation, deployment, and production impact.
- Technical depth in foundation models, large language models (LLMs), generative AI evaluation, multimodal learning, natural language processing (NLP), self-supervised learning, and computer vision using Python, TensorFlow, PyTorch, etc.
- Proven track record of partnering with clinical, business, product, and research stakeholders to align cross-functional teams, translate technical concepts for different domain experts, and deliver solutions with measurable impact.
- Experience with machine learning workflows including data processing, data curation, training pipelines, benchmarking, model evaluation, deployment collaboration, debugging, clinician-in-the-loop validation, and responsible AI review.

Experience

Senior AI/ML Engineer

September 2024 – Present

UnitedHealth Group – Optum AI

Bellevue, WA

- Led as the primary individual contributor and scientist on an autoregressive healthcare foundation model by retraining an open-source Transformer architecture on a custom Current Procedural Terminology (CPT) code vocabulary for next-service prediction; deployed as the core recommendation engine for web and mobile experiences serving up to 30 million members with sub-150 ms latency.
- Tech-led and individually contributed to 5 clinician-facing healthcare AI solutions that generate medical-record-grounded text for human-in-the-loop patient communication; built an agentic model evaluation pipeline for clinical audits using statistical NLP metrics and LLM-as-a-judge evaluation, contributing to more than \$10 million in annualized savings.
- Co-led longitudinal machine learning research across service, diagnosis, and prescription code systems, improving chronic disease prediction on MIMIC by up to 5 AUC points and improving care-variation and care-planning prediction by 5 and 8 points on Recall at 3 and Recall at 5; this foundation-model and prediction work resulted in 8 approved patent filings, 2 papers under review, and the 2025 Optum Tech Make IT Happen award.
- Built an agentic Responsible AI checker for internal audit and review teams to triage pre-deployment AI systems against policy, best practices, and historical review decisions, reducing review-cycle friction and accelerating deployment readiness.

Postdoctoral Researcher

August 2023 – September 2024

University of Washington

Seattle, WA

- Developed an NLP-based case-finding algorithm with 98% accuracy to identify specific CT imaging sequences in Python.
- Built a multithreaded deep learning data processing pipeline to measure body composition from CT images, achieving an 8x speedup.
- Partnered with the Department of Radiology to source, query, and curate research data across School of Medicine databases using SQL, supporting translational imaging studies.

Graduate Researcher

November 2018 – May 2023

University of Hawaii Cancer Center – Artificial Intelligence Precision Health Institute

Honolulu, HI

- Designed a custom biophysics-informed loss function that improved generative AI image quality by 16%.
- Created image embeddings for whole-body X-ray images using self-supervised learning with inpainting, improving downstream task performance by 15%.
- Implemented multi-view parallel learning networks to improve breast cancer risk prediction AUCs by 10% compared to previously published models.
- Wrote a spectral decomposition and convex hull clustering function to engineer imaging features that led to cancer prediction model improvement of 17% AUC.

Machine Learning Pathway Intern

May 2020 – August 2020

National Aeronautics and Space Administration (NASA)

Houston, TX

- Used shape and appearance modeling to fuse exterior 3D body scans with X-ray scans to visualize joint rotation centers and improve spacesuit ergonomics.
- Wrote and tested a Python implementation of a local system transpose to standardize 3D model point clouds.

Research Scientist

August 2015 – October 2018

EyeGenix LLC

Honolulu, HI

- Performed data-driven protein polymer design resulting in successful artificial cornea implantation in animal trials.
- Learned to rapidly prototype, test, and iterate while managing stakeholder expectations at a fast-paced startup.

Education

PhD - Bioengineering

2023

Advisor: John Shepherd, PhD

GPA: 3.95/4.00

- Dissertation: *Reducing the Burden of Cancer with Artificial Intelligence*

MS - Computer Science

2018

Advisor: Henri Casanova, PhD

GPA: 3.89/4.00

- Thesis: *A Heuristic for Optimizing the Physical Layout and Network Topology of Integrated 3D Multi-chip Systems Under Temperature Constraints*

BS - Biology

2014

Minor: Chemistry

GPA: 3.89/4.00

Fellowships

Google Computer Science Research Mentorship Program

2023

NIH AIM-AHEAD

2023

Awards

Make IT Happen - Optum Tech

2025

Merit Based Awards for Research - University of Hawaii

2023

Outstanding Dissertation Award - University of Hawaii

2023

Trainee Travel Grant - University of Hawaii Cancer Center

2019 – 2021

Graduate Research Grant - University of Hawaii

2020

Notable Publications

- [1] **Leong, Lambert T**, Michael C Wong, Yong E Liu, Yannik Glaser, Brandon K Quon, Nisa N Kelly, Devon Cataldi, Peter Sadowski, Steven B Heymsfield, and John A Shepherd. Generative deep learning furthers the understanding of local distributions of fat and muscle on body shape and health using 3d surface scans. *Communications Medicine*, 4(1), 2024.
- [2] Yannik Glaser, John Shepherd, **Leong, Lambert**, Thomas Wolfgruber, Li-Yung Lui, Peter Sadowski, and Steven R Cummings. Deep learning predicts all-cause mortality from longitudinal total-body dxa imaging. *Communications medicine*, 2(1):1–12, 2022.
- [3] **Leong Lambert T**, Serghei Malkov, Karen Drukker, Bethany L Niell, Peter Sadowski, Thomas Wolfgruber, Heather I Greenwood, Bonnie N Joe, Karla Kerlikowske, Maryellen L Giger, et al. Dual-energy three-compartment breast imaging for compositional biomarkers to improve detection of malignant lesions. *Communications Medicine*, 1(1):1–11, 2021.
- [4] Xun Zhu, Thomas K Wolfgruber, **Leong Lambert**, Matthew Jensen, Christopher Scott, Stacey Winham, Peter Sadowski, Celine Vachon, Karla Kerlikowske, and John A Shepherd. Deep learning predicts interval and screening-detected cancer from screening mammograms: a case-case-control study in 6369 women. *Radiology*, 301(3), 2021.
- [5] **Leong Lambert T** and Sean Wiere. Digit recognition from wrist movements and security concerns with smart wrist wearable iot devices. In *53rd Hawaii International Conference on System Sciences*. HICSS, 2020.
- [6] Michihiro Koibuchi, **Leong Lambert**, Tomohiro Totoki, Naoya Niwa, Hiroki Matsutani, Hideharu Amano, and Henri Casanova. Sparse 3-d nocs with inductive coupling. In *2019 56th ACM/IEEE Design Automation Conference (DAC)*, pages 1–6. IEEE, 2019.